# Math of PE, Masked Attention, and Cross-Attention

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# Sinusoidal Positional Encoding (PE)

For position $p$, model width $d_{\text{model}}$, and channel index $i$:

$$PE(p, 2i) = \sin\left(\frac{p}{10000^{2i/d_{\text{model}}}}\right),$$
$$PE(p, 2i+1) = \cos\left(\frac{p}{10000^{2i/d_{\text{model}}}}\right).$$

$$X_{\text{input}} = E + PE$$

- $E \in \mathbb{R}^{n \times d_{\text{model}}}$: token embeddings.
- $PE \in \mathbb{R}^{n \times d_{\text{model}}}$: deterministic positional signal.
- Addition injects token order before attention is applied.

# Masked Self-Attention (Decoder)

Let $Q = XW_Q,\ K = XW_K,\ V = XW_V$, where $X \in \mathbb{R}^{n \times d_{\text{model}}}$.

$$S = \frac{QK^\top}{\sqrt{d_k}} + M$$

$$A = \mathrm{softmax}(S), \qquad Z = AV$$

## Causal mask $M$

$$M_{ij} = \begin{cases} 0, & j \le i \\ -\infty, & j > i \end{cases}$$

- Ensures token $i$ cannot attend to future tokens $j > i$.
- Keeps autoregressive generation valid.

# Cross-Attention (Decoder to Encoder)

Decoder states $Y \in \mathbb{R}^{m \times d_{\text{model}}}$, encoder outputs $H \in \mathbb{R}^{n \times d_{\text{model}}}$:

$$Q = YW_Q, \qquad K = HW_K, \qquad V = HW_V$$

$$\text{CrossAttn}(Y, H) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

- Queries come from decoder ($Y$).
- Keys/Values come from encoder ($H$).
- Output rows mix source-side information relevant to each target token.

# Quick Shape Check

$$Q \in \mathbb{R}^{m \times d_k}, \quad K \in \mathbb{R}^{n \times d_k}, \quad V \in \mathbb{R}^{n \times d_v}$$
$$QK^\top \in \mathbb{R}^{m \times n}, \quad A \in \mathbb{R}^{m \times n}, \quad Z \in \mathbb{R}^{m \times d_v}$$

- In masked self-attention: $m = n$, with triangular mask.
- In cross-attention: $m \neq n$ is allowed (target length differs from source length).

**Core formula (both cases):** $\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \mathrm{mask}\right) V$

# Worked Math: Masked Self-Attention (Toy)

Take $d_k = 2$, one query row at step $i = 2$:

$$q_2 = [1, 1],\ k_1 = [1, 0],\ k_2 = [0, 1],\ k_3 = [1, 1]$$

Raw scores:

$$[q_2 k_1^\top,\ q_2 k_2^\top,\ q_2 k_3^\top] = [1, 1, 2]$$

Scale by $\sqrt{2}$:

$$s = [0.71, 0.71, 1.41]$$

Causal mask at $i = 2$: token 3 is future, so

$$s_{\text{masked}} = [0.71, 0.71, -\infty]$$

$$a = \mathrm{softmax}(s_{\text{masked}}) = [0.5, 0.5, 0]$$

If $v_1 = [2, 0],\ v_2 = [0, 2],\ v_3 = [9, 9]$, then

$$z_2 = 0.5 v_1 + 0.5 v_2 + 0 \cdot v_3 = [1, 1]$$

# Worked Math: Cross-Attention (Toy)

One decoder query $q = [1, 2]$, two encoder keys:

$$k_1 = [1, 0], \quad k_2 = [0, 1], \quad d_k = 2$$

Scores:

$$[qk_1^\top, qk_2^\top] = [1, 2], \quad s = \frac{[1, 2]}{\sqrt{2}} = [0.71, 1.41]$$

$$a = \mathrm{softmax}(s) \approx [0.33, 0.67]$$

Encoder values:

$$v_1 = [3, 1], \quad v_2 = [1, 3]$$

Output:

$$z = 0.33\, v_1 + 0.67\, v_2 \approx [1.66, 2.34]$$

- Interpretation: decoder token attends more to source token 2.

# Worked Math: Multi-Dimensional Matrix Example

Let $m = 2,\ n = 3,\ d_k = 2,\ d_v = 2$:

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad V = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$$

1) Score matrix:

$$QK^\top = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

2) Scale by $\sqrt{2}$:

$$S = \frac{QK^\top}{\sqrt{2}} \approx \begin{bmatrix} 0.71 & 0 & 0.71 \\ 0 & 0.71 & 0.71 \end{bmatrix}$$

3) Row-wise softmax:

$$A = \mathrm{softmax}(S) \approx \begin{bmatrix} 0.40 & 0.20 & 0.40 \\ 0.20 & 0.40 & 0.40 \end{bmatrix}$$

4) Output:

$$Z = AV \approx \begin{bmatrix} 1.20 & 0.80 \\ 0.80 & 1.20 \end{bmatrix}$$

$$(2 \times 2) \cdot (2 \times 3) \to (2 \times 3), \quad (2 \times 3) \cdot (3 \times 2) \to (2 \times 2)$$

# Full Process Math: PE to Final Decoder Output

**1) Source embedding + PE**

$$X^{(0)} = E_{\text{src}} + PE_{\text{src}}$$

**2) Encoder block (repeat $L_e$ times)**

$$\tilde{X} = \mathrm{LN}(X + \mathrm{MHA}(X, X, X)), \quad X' = \mathrm{LN}\Big(\tilde{X} + \mathrm{FFN}(\tilde{X})\Big)$$

$$H = X^{(L_e)}$$

**3) Target embedding + shifted input + PE**

$$Y^{(0)} = E_{\text{tgt,shifted}} + PE_{\text{tgt}}$$

**4) Decoder block (repeat $L_d$ times)**

$$\tilde{Y} = \mathrm{LN}(Y + \mathrm{MHA}_{\text{mask}}(Y, Y, Y))$$

$$\hat{Y} = \mathrm{LN}\Big(\tilde{Y} + \mathrm{MHA}_{\text{cross}}(\tilde{Y}, H, H)\Big)$$

$$Y' = \mathrm{LN}\Big(\hat{Y} + \mathrm{FFN}(\hat{Y})\Big), \quad Y^{(L_d)} = Y'$$

**5) Token logits and probabilities**

$$\text{logits} = Y^{(L_d)} W_o + b_o, \qquad P_t = \mathrm{softmax}(\text{logits}_t)$$

$$\hat{y}_t = \arg\max P_t$$

# Worked End-to-End Mini Example (One Decoder Step)

**Given:** $d_{\text{model}} = 2,\ d_k = 2$, source length $n = 2$, current target step $t = 2$.

$$E_{\text{src}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},\ PE_{\text{src}} = PE_{\text{src, simple}} \Rightarrow X^{(0)} = \begin{bmatrix} 1.10 & 1.00 \\ 0.20 & 1.98 \end{bmatrix}$$

Assume encoder output (after self-attn + FFN stacks):

$$H = \begin{bmatrix} 1.0 & 0.5 \\ 0.3 & 1.2 \end{bmatrix}$$

Current decoder state after masked self-attention:

$$\tilde{y}_2 = [1, 1]$$

Cross-attention with $q = \tilde{y}_2,\ K = H,\ V = H$:

$$qK^\top = [1.5, 1.5], \quad s = \frac{[1.5, 1.5]}{\sqrt{2}} = [1.06, 1.06]$$

$$a = \text{softmax}(s) = [0.5, 0.5], \quad z_2 = 0.5[1.0, 0.5] + 0.5[0.3, 1.2] = [0.65, 0.85]$$

Final linear to vocab size 3, $W_o = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix},\ b_o = 0$:

$$\text{logits}_2 = z_2 W_o = [0.65,\ 0.85,\ -0.20]$$

$$P_2 = \text{softmax}(\text{logits}_2) \approx [0.38,\ 0.46,\ 0.16]$$

$$\hat{y}_2 = \arg\max P_2 = \text{token } 2$$

# How $PE_{\text{src}}$ Comes from Sin/Cos

For $d_{\text{model}} = 2$, channels are:

$$PE(p, 0) = \sin(\theta_p), \qquad PE(p, 1) = \cos(\theta_p), \qquad \theta_p = \frac{p}{10}$$

For positions $p = 1, 2$:

$$PE(1, :) = [\sin(0.1), \cos(0.1)] \approx [0.0998, 0.995]$$

$$PE(2, :) = [\sin(0.2), \cos(0.2)] \approx [0.1987, 0.980]$$

$$PE_{\text{src, exact}} \approx \begin{bmatrix} 0.0998 & 0.995 \\ 0.1987 & 0.980 \end{bmatrix}$$

Simplified classroom PE (used in worked hand math): $PE_{\text{src, simple}} = \begin{bmatrix} 0.10 & 1.00 \\ 0.20 & 0.98 \end{bmatrix}$